Tuberculosis Infection Prevention and Management Guideline

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All areas	All workforce groups

2. Purpose

The purpose of this policy is to provide guidance on the infection prevention and management of patients with known or suspected Tuberculosis (TB) and identification of TB contacts in the healthcare setting. It is designed to ensure the risk to patients, staff and visitors are minimised. For details on pre-employment screening of TB for healthcare workers, refer to the AKG Healthcare Worker Immunisation and Health Policy. Medical management of patients with TB is outside the scope of this guideline.

3. Background

Tuberculosis (TB) remains a leading cause of death worldwide. It is present in all countries and age groups, is curable and preventable[2]. TB is an infectious disease caused by the bacteria, *Mycobacterium tuberculosis*.

TB is classified as pulmonary or extra-pulmonary[1].

Pulmonary TB is more common and refers to disease involving the lung parenchyma and the trachea or bronchi.

Extra-pulmonary TB is disease involving any other part of the body and includes lymph node, pleural space, skeletal, urogenital and disseminated TB.

NOTE: TB is a Notifiable Disease under the Health Act of Western Australia.

4. Early Detection

Early detection of pulmonary TB is essential and will reduce the risk of transmission to patients and staff. The likelihood of TB should be assessed based on the clinical presentation and risk factors for TB.

4.1 Clinical presentation

Approximately two thirds of all cases in Australia present with pulmonary TB (disease involving the lungs) and can have the following common symptoms[3].

- Fever and night sweats
- Unintended loss of weight
- Feeling generally tired and unwell
- Cough of any duration but especially lasting greater than 2 weeks
- Haemoptysis

Symptom onset may be more acute in children[3]. Clinical suspicion of TB should be high in any person with possible TB exposure and a respiratory infection unresponsive to standard treatments, or an unexplained non-respiratory illness.

Extra-pulmonary TB can present with a wide range of symptoms and is often accompanied by intermittent fever or weigh loss.

4.2 Persons at increased risk

The most important risk factor for TB in a person with relevant symptoms is the prior risk of exposure to TB. In Western Australia (WA) the main risk factor is birth or prior residence in a country with high TB incidence (>40/100 000 per year). For country-based TB incidences, refer to the World Health Organization (WHO) interactive site . About half of TB cases in WA are diagnosed in people within 4 years of migration from a high TB incidence country[1].

Other important risk factors for TB infection include a history of contact with another person with TB, being a HCW, being an Aboriginal Australian or being born prior to 1950[1].

Some groups are more susceptible to progression to TB disease following infection. High risk groups include[3].

- Children <5 years of age
- Elderly people
- People with severe malnourishment
- People living with HIV, poorly controlled diabetes, renal failure, or any immunocompromising disease
- Patients receiving immunomodulating therapies, e.g. high dose corticosteroids or cancer chemotherapy

5. AHS management of patients with suspected or confirmed pulmonary TB

For Kalamunda Hospital, refer to Section 6.

5.1 Notification of suspected or confirmed cases

Medical officer to ensure timely referral to:

Inpatients – AKG Infectious Diseases via switch

Outpatients – The Anita Clayton Centre (ACC), (08) 9222 8500

5.2 Isolation and Precautions

Patients diagnosed with suspected or confirmed pulmonary TB must be isolated in a negative pressure isolation room (NPIR) under airborne precautions. The use of a portable air purifier unit within a standard single room is not considered to provide the equivalent transmission risk mitigation as an NPIR.

Investigations that may produce infectious aerosols, such as sputum collection and aerosol therapy, should be deferred where possible until the patient is admitted into a ward NPIR under airborne precautions.

Patients should be requested to wear a surgical mask until placed into an NPIR.

Negative Pressure Isolation Rooms are located:

- Emergency Department, Bay 46
- Canning Ward, Room 2
- ICU, Room 3

5.3 Personal Protective Equipment

Before entering the room of a patient with suspected or confirmed pulmonary TB, all HCWs must wear a P2 or N95 particulate filter respirator (PFR) that they have been fit-tested and trained in its safe use, this includes performing a fit check every time a PFR is donned.

HCWs with known immunosuppression should not care for the patient.

5.4 Specimens

As *Mycobacterium tuberculosis* may be present intermittently in sputum, 3 separate early morning sputum specimens are ideally collected on consecutive days, or an induced sputum collected by a physiotherapist is required.

Due to the slow growth rate of mycobacteria, microscopy for acid-fast bacilli (AFB) will be performed with results available within 24 hours. If the specimens under suspicion for TB have AFBs present, the patient is considered ‘smear positive’. Culture will also be performed; these specimens can take up to 4 weeks for results.

Those patients with TB who are smear positive (AFBs detected) are considered more infectious than those who are smear negative.

In the hospital setting, smear negative, culture positive pulmonary TB cases are to be managed as smear-positive pulmonary TB patients, including clearance criteria under the direction of the Clinical Microbiologist/Infectious Diseases physician.

5.5 Visitors

Visitors must generally be restricted to family members and/or household contacts of the patient. Children under the age of 15 years, other than household contacts and/or direct family members should be excluded from source isolation rooms. In general, visitors must wear a fit-*checked* P2 or N95 PFR while the patient remains under isolation.

5.6 Removal of precautions

Suspected cases should remain isolated under airborne precautions until negative results of 3 consecutive sputum smear specimens are available.

Generally, patients will be considered non-infectious after 14 days of appropriate antimicrobial therapy. The decision to remove precautions must be made in consultation with the Clinical Microbiologist/Infectious Diseases physician.

IP&M **must** be informed before removing precautions from a smear positive patient.

5.7 Discharge

The room must be closed for 1 hour **prior** to cleaning.

Patient Support Services (PSS) to clean the room in accordance with 'Isolation Room Discharge' procedure in the AKG Cleaners Manual.

5.8 Management of extra-pulmonary TB cases

These patients usually do not require special precautions except those with laryngeal TB, or TB skin abscesses or ulcers that require dressings or irrigation.

6. KH management of patients with suspected or confirmed pulmonary TB

Inform treating Physician, Infection Prevention and Management and Nurse Manager

- IP&M to contact AKG ID physician or, if not available, an ACC respiratory physician to facilitate transfer to appropriate facility for treatment & management
- Whilst awaiting transfer, patient to wear a surgical mask. If there may be a delay with the transfer, move the patient to a single room and keep the door closed
- Staff must wear appropriate PPE as above
- The ambulance service should be informed of the patient's diagnosis, and the patient should wear a surgical mask during transfer
- Investigations and treatment that may induce coughing must be deferred where possible, until active TB has been excluded, or the patient is in a negative pressure isolation room at another facility
- After discharge, clean the room as per isolation room discharge cleaning process. The room must be **closed** for 1 hour **prior** to cleaning

7. Special populations

7.1 Paediatrics

Clinical, diagnostic and management differences between TB in children and TB in adolescents/adults include the following:

- Children are at higher risk of TB disease following primary infection compared with adults, especially in very young (<5 years) and immunocompromised children.
- Children <5 years are at higher risk of developing severe disseminated forms of TB (e.g., miliary and meningeal TB) compared with adults.
- Diagnosis of active TB in children should be based on the exposure risk and clinical/radiological signs of TB. Microbiological confirmation of disease is not always found owing to the paucibacillary nature of disease. Negative molecular diagnostics and/or mycobacterial culture on diagnostic samples should not be used to exclude TB in a child.

- TB in children can be difficult to diagnose because the symptoms and signs are often non-specific, children are often unable to express their symptoms, and diagnostic confirmation of disease is often not achieved. The most important clue to suggest TB as a possible diagnosis is recent exposure to an infectious TB case

7.1.1 Clinical presentation

Symptoms in children suggestive of TB disease include:

- A persistent cough, often non-productive, >2 weeks' duration
- Weight loss or failure to thrive
- Reduced playfulness, fatigue or increased tiredness
- Enlarged, non-tender cervical lymph nodes (>2cm x 2cm) with/without overlying skin changes or fistulae.

7.1.2 Respiratory samples

Respiratory samples should be collected in any child or adolescents with symptoms concerning for active TB and/or with abnormal CXR findings and a positive screening test.

A minimum of two **early morning** respiratory samples are ideally obtained. It may be pragmatic to collect both samples on the same day. Sampling types include:

- Induced sputa
  - Performed with the assistance of a respiratory physiotherapist in children >3 months of age
- Gastric aspirates
  - In young children, sputum is swallowed rather than expectorated. Consecutive early morning gastric aspirates obtained via nasogastric or orogastric tube may be collected as an alternative to induced sputa. Early morning gastric aspirates need to be obtained while a child is fasting and before a child is ambulant.
- Bronchoalveolar lavage (BAL)
  - BAL samples may be indicated in specific situations, however BALs should be performed by an experienced operator with appropriate infection control precautions, ideally at the end of a procedural list.
- Stool samples
  - Stool samples may be considered for AFB and mycobacterial culture.

7.1.3 Perinatal TB

Perinatal TB encompasses TB acquired by the baby while i) in-utero, ii) intrapartum or iii) during the postnatal period. All pregnant women with TB should be discussed with the Perth Children's Hospital (PCH) paediatric on-call registrar/consultant to allow appropriate risk stratification and management of the mother and infant. The registrar can be contacted via the PCH switchboard on (08) 6456 2222

8. Exposure management

When a patient or HCW is diagnosed with TB, follow up of other HCW and patients that have had contact, or exposure to the index case may be necessary. Each exposure will be reviewed on a case-by-case basis to determine if "significant exposure" has occurred.

Significant exposure is defined as:

- Shared airspace, on a single occasion or cumulatively, for more than 8 hours.
- Contact involving a procedure that confers increased risk (e.g. sputum induction, bronchoscopy, intubation, post-mortem examination) where airborne precautions had not been implemented; and/or
- Contact where physical containment requirements in a microbiological laboratory are breached.

8.1 Management of patients following exposure to TB

- IP&M staff review all confirmed cases of pulmonary TB to determine if follow up is required, in consultation with the Clinical Microbiologist/Infectious Diseases physician and the ACC case manager.
- Inpatients who have had significant exposure to a smear positive pulmonary TB (e.g. shared a room with the source case for ≥8 hours and room contacts of a culture positive, smear negative pulmonary TB patient) require follow-up.
- A letter will be sent to these contact patients as well as their General Practitioners informing them of the exposure. The letter will outline the symptoms of TB and direct the patient to seek medical advice if these symptoms are experienced over a prolonged period.

8.2 Management of HCWs following exposure to TB

- HCWs who have significant contact with an index case are considered casual contacts
- Those identified will be notified in writing of potential exposure as soon as possible. The information will include an outline of the symptoms of TB and options for follow-up.
- Casual contacts generally only need one test after 8-12 weeks
- HCWs who have completed post exposure screening and their QuantiFERON-TB is positive, or were previously positive from a pre-employment screening will be referred to ACC for review and management.

9. Compliance monitoring

Compliance against this guideline will be monitored by the Infection Prevention and Management Committee. Monitoring activities will include:

- Infection Prevention and Management surveillance reporting
- Clinical incident reported data via the Datix Clinical Incident Management System (CIMS).

10. Legislative context

The following documents are required to give effect to this policy:

- *Health Services Act 2016*
- *Public Health Act 2016*
- *Public Health Regulations 2017*

11. Supporting information

The following documents support the implementation of this policy:

- EMHS
    - Infection Prevention and Management Policy (EMHS: 34)
- AKG
    - Hospital Outbreak Management Policy
    - Healthcare Worker Immunisation and Health Policy
    - Standard and Transmission Based Precautions Procedure
    - Patient Support Services Cleaners Manual

12. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.08
- Standard 3 – Action 3.16

13. References

1. Public Health and Clinical Excellence. (2025, July). *Guidelines for Tuberculosis Control in Western Australia: Western Australian Tuberculosis Control Program*. Government of Western Australia. https://www.health.wa.gov.au/-/media/Files/Corporate/general-documents/Tuberculosis/PDF/WATBCP-Guidelines-for-TB-Control.pdf
2. World Health Organization [WHO]. (2025). *Tuberculosis*. https://www.who.int/news-room/fact-sheets/detail/tuberculosis
3. Communicable Disease Network Australia [CDNA]. (2022, March). *Tuberculosis. CDNA National Guidelines for Public Health Units.* Australian Government. https://www.health.gov.au/sites/default/files/documents/2022/06/tuberculosis-cdna-national-guidelines-for-public-health-units.pdf

14. Glossary

The following definitions are relevant to this policy.

Term	Definition
Acid-fast bacilli (AFB)	Mycobacteria, including *Mycobacterium tuberculosis* which causes TB, that can be seen under the microscope following a staining procedure in which the bacteria retain the colour of the stain after an acid wash (acid-fast).
AFB smear testing	A microscopic examination of sputum that is stained to detect acid-fast bacteria. It is a rapid test used to provide presumptive results within one to two days. AFB smears must be confirmed with AFB culture.
AFB culture	Mycobacterial culture remains the gold standard for the definitive diagnosis of TB disease, however positive results can take several weeks, while negative culture results can take up to 6-8 weeks.
Anita Clayton Centre (ACC)	WA TB Control Program (contact 9222 8500 or ACCadmin@health.wa.gov.au)
Negative Pressure Isolation Room (NPIR)	A room in which the air pressure differential between the room and the adjacent indoor airspace directs the air flowing into the room i.e. room air is prevented from leaking out of the room and into adjacent areas such as the corridor.
Particulate Filter Respirator (PFR)	Respirators that filter at least 94 percent of 0.3-micron particles from the air. PFRs are used when implementing airborne precautions. Both P2 and N95 respirators are appropriate for use with airborne precautions.
QuantiFERON-TB	A blood test used to detect TBI. A positive QFT test suggests TBI, it does not indicate the presence or absence of TB disease.
Smear-positive pulmonary TB	A diagnosis of pulmonary TB based on culture of *Mycobacterium tuberculosis* in which AFBs have been seen on sputum microscopy.
Smear-negative, culture positive pulmonary TB	A diagnosis of pulmonary TB based on culture of *Mycobacterium tuberculosis* in which AFBs have NOT been seen on sputum microscopy.
TB infection (TBI)	A state of persistent immune response to stimulation by *Mycobacterium tuberculosis* antigens without evidence of TB disease. Formerly known as latent tuberculosis infection (LTBI).
TB disease	An airborne bacterial infection caused by the organism *Mycobacterium tuberculosis* that primarily affects the lungs, but can involve the kidneys, bone, spine, brain and other parts of the body.

15. Policy Contact

Director Nursing and Midwifery

Enquiries relating to this guideline may be directed to:

Title: Clinical Nurse Specialist
Department: Infection Prevention and Management
Email: ahsinfection.controlnurses@health.wa.gov.au

16. Review

This policy will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	24 11 2025	24 11 2028	Full review; updated to align with WA Guidelines and addition of management of children and woman.

17. Authorisation

This policy has been authorised and issued by:

Authorised by	Russell Colyer- Cockburn - Director of Nursing and Midwifery
Authorisation date	24 11 2025
Published date	24 11 2025